1750–2024

Global CO_2 Emissions

Trends, Regional Shifts, and Per Capita Inequalities

A data-forward briefing for policy analysts and climate negotiators

CO_2

Scope: atmospheric concentration, total emissions, regional shares, per capita inequalities, and 2024 change.

>35 Bt
annual fossil fuel & industry CO_2 today
Growth has slowed, but not peaked

429 ppm
atmospheric CO_2 at Mauna Loa
February 2026 measurement

150×
per capita gap between high and low emitters
US/Australia vs. Chad/Niger

Sources: NASA; Our World in Data — CO_2 and Greenhouse Gas Emissions; Global Carbon Budget 2025.

GLOBAL CO_2 EMISSIONS

Atmospheric CO_2 Reaches 429 ppm — 50% Above Pre-Industrial Levels

429 ppm

NOAA Mauna Loa Observatory

Latest measurement: February 2026

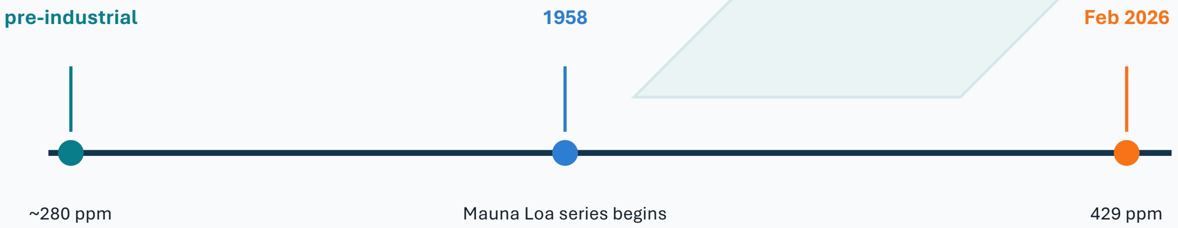
~280 ppm

pre-industrial baseline

Reference level in source brief

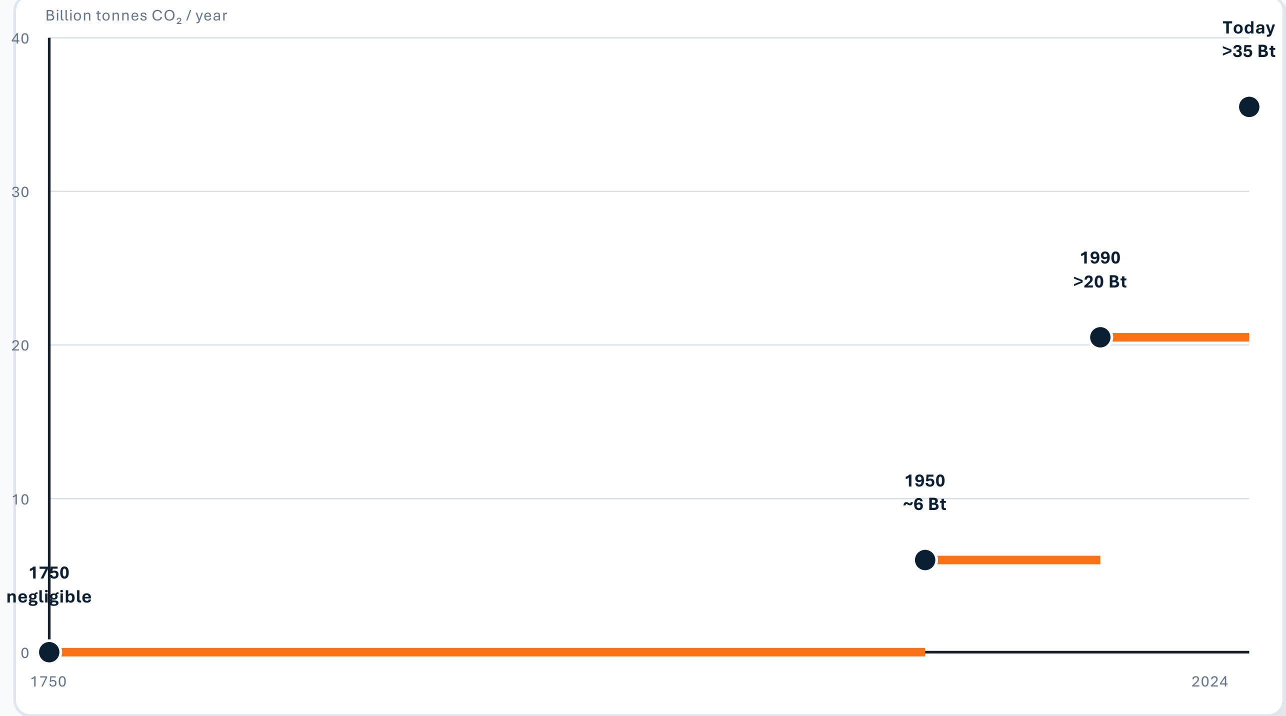
Current atmospheric CO_2 is more than 50% higher than the pre-industrial baseline. Ground-based and satellite measurements confirm a sharp acceleration linked primarily to coal, oil, and natural gas combustion.

The same magnitude of natural change took thousands of years after the last ice age; this rise occurred in roughly two centuries.



GLOBAL CO_2 EMISSIONS

Global Fossil Fuel CO_2 Emissions Grew Six-Fold: 6 Billion to 35+ Billion Tonnes



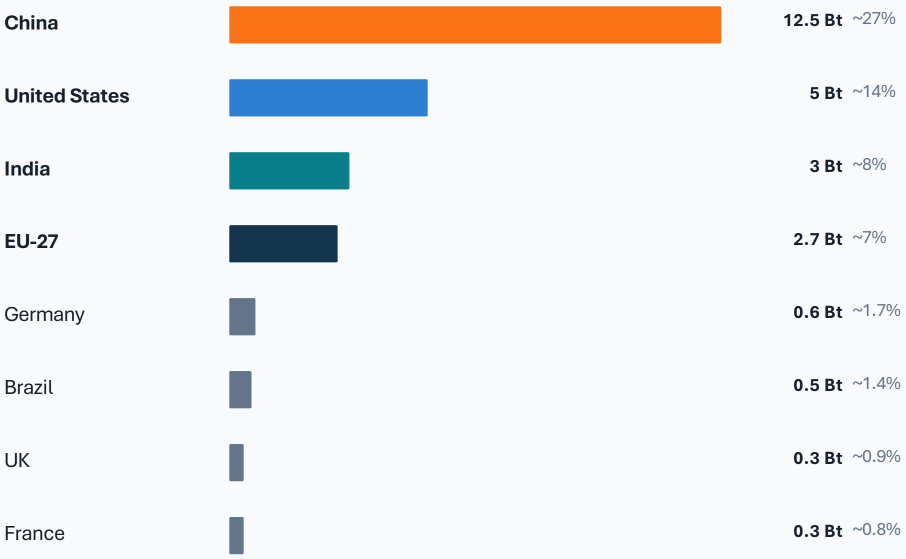
The brief anchors the trajectory at negligible pre-industrial emissions, ~6 Bt in 1950, >20 Bt by 1990, and >35 Bt today. Growth has slowed recently, but the global total has not yet peaked.

Figure 2 note

The specified figure_2.jpg was not included in the asset folder; this schematic uses only values provided in source_brief.md.

Sources: NASA; Our World in Data / Global Carbon Budget 2025; source brief.

China Now Emits Over 12 Billion Tonnes — More Than Double the United States



China produces ~12.5 Bt annually — over one-quarter of the global total and more than double the United States. The US peaked near 6 Bt around 2005 and has declined to roughly 5 Bt; India has risen rapidly to ~3 Bt.

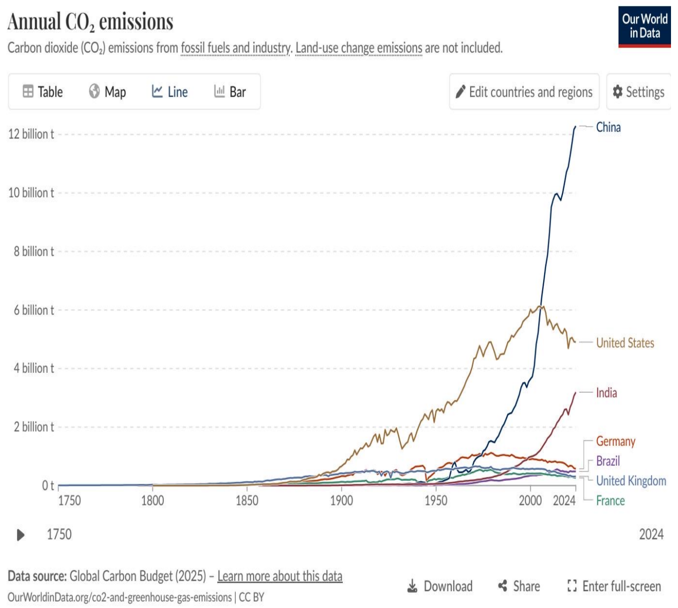
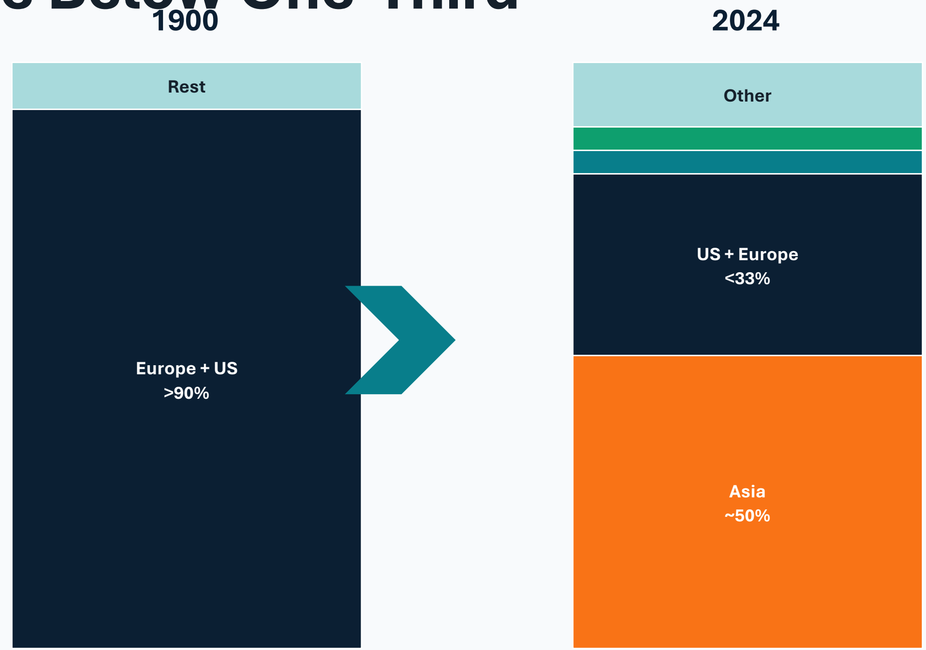


Figure 3: Annual CO_2 emissions by country

GLOBAL CO_2 EMISSIONS

Asia Now Accounts for ~50% of Global Emissions; US + Europe Below One-Third



>90%

Europe + US share in 1900

By 1950: still over 85%

~50%

Asia share today

Asia has nearly 60% of population

3–4%

Africa and South America each

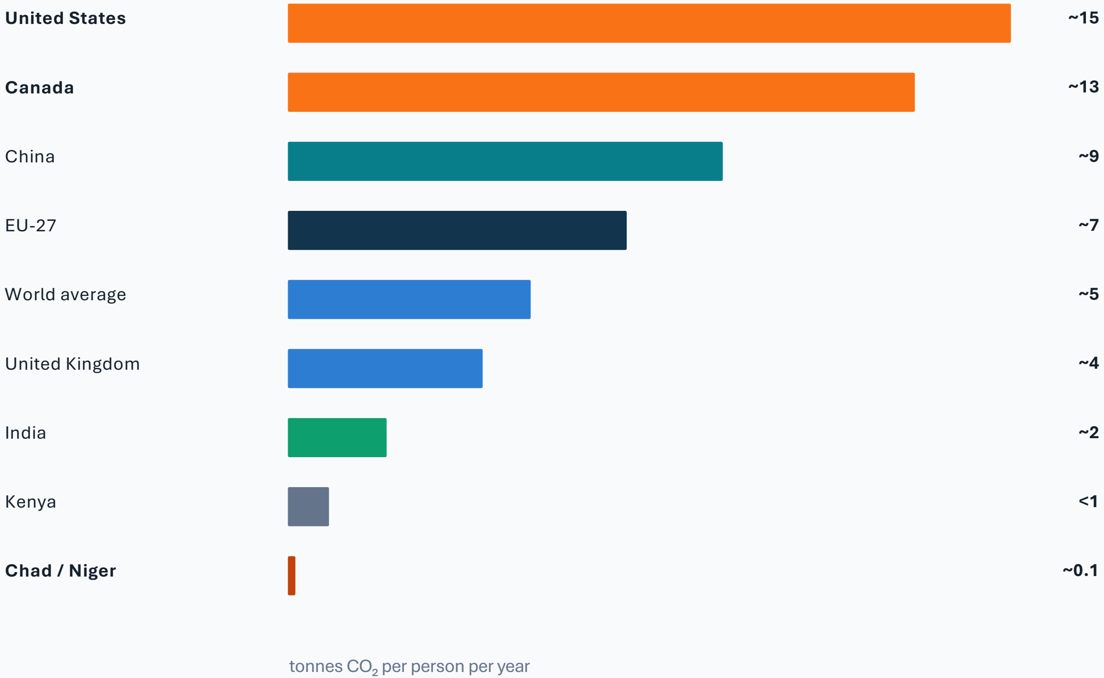
Comparable to aviation + shipping

Visual: side-by-side comparison of regional emission shares, showing the shift from Europe + US dominance in 1900 to Asia as the largest current contributor.

Sources: NASA; Our World in Data / Global Carbon Budget 2025; source brief.

GLOBAL CO_2 EMISSIONS

Per Capita Gap: 15 t in the US vs. 0.1 t in Chad — a 150× Disparity



150×

approximate gap between US/Australia and Chad/Niger

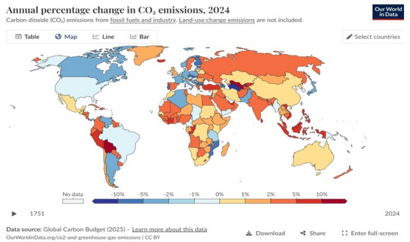


Figure 1: per capita CO_2 emissions over time

The average American or Australian emits in under two days what the average person in Chad or Niger emits in an entire year.

Sources: NASA; Our World in Data / Global Carbon Budget 2025; source brief.

2024 Snapshot: US and Europe Cut Emissions While Russia and SE Asia Increase

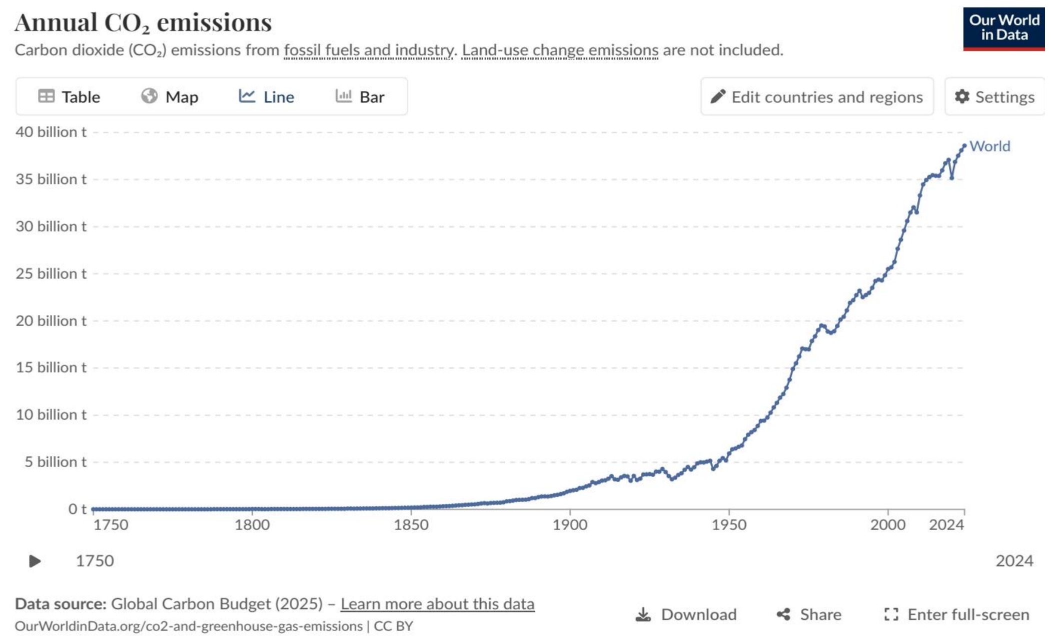


Figure 4: Annual percentage change in CO_2 emissions, 2024

Declines

United States and much of Europe recorded negative change, roughly −1% to −5%, shown in blue on the source map.

Increases

Russia, parts of Southeast Asia, and several African nations saw increases of +2% to +10% or more, shown in orange/red.

South America shows mixed outcomes, including one or two very large percentage increases (>10%).

Sources: NASA; Our World in Data / Global Carbon Budget 2025; source brief.

Energy Mix Explains Why France Emits Far Less Per Capita Than Germany

Countries with similar living standards can have very different per capita emissions. The source brief attributes lower France/UK emissions largely to higher nuclear and renewable shares, while Germany/Netherlands have higher fossil fuel reliance.

Country set	Energy sources noted in brief	Per capita CO_2	Interpretation
France / UK	Higher nuclear and renewable shares	France ~4 t; UK ~4 t	Lower per capita emissions among high-income countries
Germany / Netherlands	Higher fossil fuel share	Higher than France / UK; exact values not specified in brief	Energy mix, not only prosperity, drives differences

~4 t
France and UK per person in source table
Both listed below EU-27 average (~7 t)

Policy
energy system choices affect outcomes
Nuclear, renewables, and fossil shares matter

Not just income
similar prosperity, different emissions
Source brief finding

Key Takeaways

Emissions are still rising, but the map of responsibility is shifting.

>35 Bt	Global CO_2 emissions exceed 35 billion tonnes per year and have not yet peaked.
>25%	China alone accounts for over one-quarter of global emissions (~12.5 Bt), more than double the US (~5 Bt).
~50%	Asia now produces roughly half of global emissions; Europe + US fell from >90% in 1900 to below one-third today.
150×	Per capita inequalities remain extreme between high emitters and Chad/Niger (~0.1 t/person).
Energy mix	Policy and energy choices — not just prosperity — determine per capita outcomes.

Negotiation implication: account for both current annual totals and durable per capita inequalities when framing responsibility, transition support, and energy-system choices.

Sources: NASA; Our World in Data / Global Carbon Budget 2025; source brief.